

Watching Live: The Master Checklist and Anticipation

Twenty seconds between snaps, nine questions in the right order, the full catalogue of tells, and a prediction with a number on it.

The NFC Championship Game, January 25, 2026. Third quarter, 8:23 to play, and the Rams have just pulled within four points. Seattle faces third-and-9 at its own 36, and on a typical broadcast the replay of the run that just lost two yards would still be rolling as the Seahawks come back to the line: one running back, one tight end, three receivers, quarterback in the shotgun. The Rams answer with six defensive backs and only five men near the ball. You have about twenty seconds. Most viewers spend them waiting for the snap. A coach in the booth spends them asking a set of questions in the same order he asks them before every snap: *What does the situation demand? Who's on the field? How are they lined up? What does the defense do when someone moves? How many are in the box? How many are deep? Where is the help? Is anyone coming?* And then he says out loud what he thinks will happen, and how sure he is. This chapter puts the whole course into that routine. By the end of it you'll run the same questions in the same order, and you'll keep score.

You'll be able to...

- Run the master pre-snap checklist in about twenty seconds, in order: situation → personnel → formation and strength → motion response → box → shell → leverage → pressure tells → prediction, and know which steps to keep when the offense goes fast.
- Use the full tells catalogue (linemen's stance and weight, the back's depth and offset, pistol against shotgun, receiver splits, the tight end's stance, linebacker depth, defensive back depth and leverage, safety width), saying for each what it usually means, why, and how often it lies.
- Say what each step of the checklist is worth, from 2024–2025 data, and why the later steps answer different questions from run or pass.
- Make layered predictions (run or pass first, then the coverage family, the rush, where the ball goes), each with a stated confidence between 50% and 90%.
- Follow a complete playoff drive snap by snap the way a coach in the booth does, and keep the log that Film Study will score.

You'll need

This is a Level 5 chapter: it assembles the course rather than adding new schemes. One line on each piece it leans on hardest:

- **In-game adjustments and tempo** (In-Game Adjustments, Tempo, and the Sideline Battle): tempo is how fast the offense gets to the line and snaps; a hurry-up offense snaps as fast as it can, and the substitution freeze means a defense that can't match a substitution has to play with the eleven it has. Staffs adjust between series, and the play that follows a successful call (the call after the call) often builds on it.
- **The clock and fourth down** (The Clock, the Two-Minute Drill, and Fourth Down): in the two-minute drill and the four-minute offense the clock, not the defense, decides much of the play call; fourth down is a decision with its own rules of thumb.
- **Tendencies from data** (Tendencies: PROE, Personnel, Motion, and Coverage from Charting Data): xpass (expected pass rate) is the chance of a pass given the situation alone, and PROE (pass rate over expected) is how much a team throws beyond it. A team's tendency is a pattern in what it calls from a given situation or look.
- **The habit** (How to Watch a Broadcast Like You Mean It): a tell is a pre-snap clue that shifts the odds, never a guarantee; the prediction log is your record of calls, with a confidence column you've been filling in since Part 1. The "checklist so far" cards at the end of Motion and Shifts, Offensive Systems, Disguise and Rotation and Zone Blitzes are the drafts of the card this chapter finishes.

Twenty seconds, a hundred clues

Start with the problem the checklist solves: there's far more to see before a snap than anyone can see, and very little time. The play clock is 40 seconds, starting the moment the previous play is whistled dead (25 seconds after some stoppages, such as a timeout

or a change of possession).¹ On a broadcast the first part of that is usually spent on replays of the play that just ended. The offense huddles while its play-caller talks to the quarterback through the helmet radio, which goes silent with 15 seconds left on the play clock.² Then the players walk to the line, set, maybe move someone in motion, and the ball is snapped. A team that huddles spends most of what is left of the play clock at the line; a team in tempo can snap within a few seconds of setting.

So your budget is roughly twenty seconds, from the moment the scorebug updates to the snap, and the camera won't show you all of it. Within those twenty seconds, information doesn't arrive all at once. It arrives in a fixed order, because the game produces it in a fixed order:

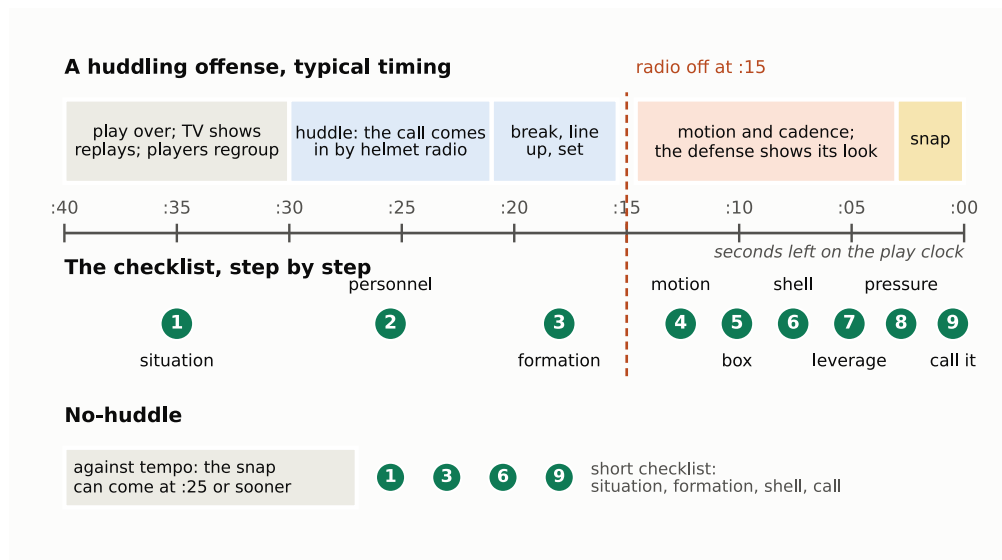


Figure 1: Twenty seconds, in order. Top: what a typical huddling offense does with the 40-second play clock (timings vary by team and snap; only the radio cut-off at :15 is a rule). Middle: the nine pre-snap steps of the master checklist, placed where each answer usually becomes visible: the situation during the replay, the personnel as players run on, the formation when the offense sets, the motion and the defense's reply during the cadence, and the box, shell, leverage and pressure in the last few seconds, when the defense finally shows its hand. Bottom: against tempo everything is compressed, so you keep only steps 1, 3, 6 and 9.

That ordering is the first reason the checklist has the shape it has: **each step is asked when its answer is visible**. There is no point hunting for the safeties while the offense is still in the huddle; they haven't settled. There is every point in reading the scorebug during the replay, because the situation is already fixed and nothing else is.

The second reason is less obvious and more important: **each answer changes what the next clue means**. A safety walking down to eight yards on third-and-9 is a defense guarding the sticks; the same move on first-and-10 against two tight ends is an eighth man for the run. A receiver tightening his split means one thing from a team that runs on 70% of its snaps in that look and another from a team that throws on 70%. Clues aren't independent bits of evidence to be added up. They are answers to questions, and the questions come in an order. Read the box before you know the personnel, and you'll miscount it. Read the shell before you know the down, and you won't know what you're looking at.

¹NFL Football Operations, "Football terms" glossary (the 40-second play clock, and the 25-second clock after certain administrative stoppages): <https://operations.nfl.com/learn-the-game/nfl-basics/terms-glossary/>

²NFL, *Official Playing Rules*, Rule 5, Section 3, Article 3 ("Speakers in helmets"): coach-to-player communication "is cut off when the play clock reaches 15 seconds or the ball is snapped, whichever occurs first." Rulebook: <https://operations.nfl.com/the-rules/nfl-rulebook/>. The rule and its history are covered in The Pre-Snap Phase.

! Why it exists: a checklist, of all things

A checklist sounds like the opposite of expertise. Surely the expert just *sees* it? But the professions that have to perform under time pressure without missing anything (pilots running a pre-flight check, surgical teams before an incision) settled long ago on short, ordered checklists, and not because their people were beginners. A checklist offloads the *remembering* so that attention is free for the *judging*.³ The football version works the same way. The quarterback's pre-snap routine (count the safeties, find the Mike, know where the hot throw goes) is a checklist; so is the defensive signal-caller's (formation, strength, the check that goes with it). Your job is easier than theirs, since nobody is about to hit you. But it has the same enemy: the eye goes to the most interesting thing on the screen (usually the quarterback) and stays there, and the clue that decided the play happens forty yards away.

What it gives up: a fixed routine is slower than a glance, and on the snaps where the offense hurries you'll have to cut it short. That's why the card has a short version.

The master checklist

Here is the card you've been building since Part 2, finished. It is the course's **master pre-snap checklist**: the complete pre-snap routine, in the same order before every snap, followed by what to check at the snap and between snaps.

³Atul Gawande, *The Checklist Manifesto: How to Get Things Right* (Metropolitan Books, 2009), on aviation checklists and the surgical safety checklist, and on the difference between "do-confirm" and "read-do" checklists; the watching checklist is the do-confirm kind, run from memory and checked against the card.

The master checklist

new in the capstone

BEFORE THE SNAP • about 20 seconds

- | | | | |
|----------|-----|---|------------------------|
| 1 | 3 s | Situation: down, distance, field zone, clock, score, timeouts. Say the base rate: “third-and-9, pass, 90%-plus”. Two-minute or four-minute? | Part 1, Part 10, 13-04 |
| 2 | 2 s | Personnel: count backs and tight ends as they run on. Did the defense match (base, nickel, dime), or did tempo freeze it? | 02-01, 02-05, 08-04 |
| 3 | 3 s | Formation and strength: under center, gun or pistol; the back's depth and offset; splits; the tight end's stance and side; any heavy or light hands | Part 2, tells |
| 4 | 2 s | Motion response: does a defender travel (man), bump (zone) or spin a safety? Still moving at the snap? | 02-06, 06-06 |
| 5 | 2 s | Box: blockers against defenders near the line. Light (6 or fewer) or loaded (8 or more)? | 02-05, 05-05 |
| 6 | 2 s | Shell: one-high or two-high; each safety's depth and width; who is creeping late? | 06-01, 06-06 |
| 7 | 2 s | Leverage: corners press or off, inside or outside; linebacker depth; where is the help? | 04-03, Part 6, tells |
| 8 | 2 s | Pressure tells: mugs, a safety or nickel walked up, a lineman on his heels. Count the deep players; where is the hot? | Part 7, 04-02, tells |
| 9 | 2 s | Call it: run or pass with a stated confidence (50–90%), plus one second-layer call (coverage, rush or target) | 15-01 |

AT THE SNAP

- | | | | |
|-----------|--|---|-----------|
| 10 | | Linemen's first step (run or pass); did a safety rotate; how many rushed? | Parts 3–7 |
|-----------|--|---|-----------|

ONE SECOND AFTER

- | | | | |
|-----------|--|--|---------------|
| 11 | | Middle of the field closed or open; corners squat, bail or trail; where did the ball go? | Part 6, 08-01 |
|-----------|--|--|---------------|

BETWEEN SNAPS

- | | | | |
|-----------|--|---|---------------------|
| 12 | | Log it: call, confidence, the clue that decided it, the result. Five seconds, every snap | 01-06, 15-01 |
| 13 | | At each stoppage: update today's tendencies (this personnel, this down); tempo and clock status; what did they just adjust? | 08-04, 10-04, 13-04 |

Figure 2: The master checklist: the cards from Parts 2, 4, 6 and 7, completed and reordered for speed. Nine pre-snap steps take about twenty seconds in all (the seconds column is a target, not a rule). The shaded lines are what this chapter adds: the full tells catalogue folded into steps 3, 7 and 8, the prediction with a stated confidence as an explicit step, and the between-snaps habits that keep your read of this team, today, up to date. The right-hand column says where each line was taught.

Compare it with the last draft, the seventeen-line card at the end of Zone Blitzes and Simulated Pressure, and three changes stand out. **It's shorter before the snap**: nine steps, not ten, because the protection question (where is the back, which way will the line slide) has been folded into the pressure step, where you need it. **The tells have a home**: instead of a separate “run tells” and “pass tells” line, each tell sits in the step where you'd see it, offensive alignment tells in step 3, defensive depth and leverage in step 7, pressure tells in step 8. And **the prediction is a step of its own, with a number**: not “I had a feeling it was a pass” after the fact, but “pass, 80%” before it.

The rest of this section walks the nine pre-snap steps in order. For each: what you're looking at, what it tells you, and why it sits where it does. None of it is new; what's new is the order and the speed.

Step 1: situation, the base rate

Down, distance, field zone, clock, score, timeouts. You learned to read the scorebug in five seconds in How to Watch a Broadcast; Part 10 taught you what each situation does to the play-caller. The output of step 1 is a number: the **base rate**, how often offenses throw in this situation before anyone has lined up. That is what xpass measures, and it's the anchor every later step moves.

You don't need a model in your head, just a few landmarks from the 2025 regular season:⁴

Situation	Pass rate, 2025	What it means for your call
All snaps	60%	"Always pass" is the floor any reader has to beat
1st-and-10	50%	A coin flip until the formation speaks
2nd-and-1 or 2	31%	Mostly runs, but this is where shots happen
3rd-and-1 or 2, under center	19%	Run, and the formation says so
3rd-and-1 or 2, shotgun	52%	A genuine coin flip
3rd-and-7 to 10	94%	Pass; spend your seconds on the second layer
No-huddle, any down	70%	Mostly offenses that are behind; under one in five comes in a two-minute drill

Two things to say alongside the number. First, **is this a clock situation?** In the two-minute drill and the four-minute offense the clock calls half the plays, and The Clock, the Two-Minute Drill, and Fourth Down gave you a separate set of questions for them (in bounds or out, timeout or spike, kneel or not). Second, **does this team break the base rate?** Some offenses throw far more than their situations suggest and some far less; that is PROE, and Tendencies shows how to look it up before the game (its one-page scouting report is the card to bring). A team with a high PROE moves your 50% first down toward 55 or 60 before you've seen a thing.

Why it comes first: it's the only clue that's complete during the replay, and it is the biggest single clue there is. Everything else adjusts it.

Step 2: personnel, and the match

Count the backs and tight ends as the offense runs on, and watch who runs *off*. The two digits (11, 12, 21, 13) were the first thing Part 2 taught, because personnel is chosen before the formation: the play-caller sends in a group, and the group limits what the formation can be. Then glance at the defense's answer. Did it stay in base, go to nickel, go to dime? A defense in nickel against 12 personnel has a defensive back where a linebacker usually stands, and the offense will often try to run at him; a defense in base against 11 personnel has a linebacker who may have to cover a slot receiver, and the offense will often try to throw at him.

This is also where you notice **tempo**. If the offense hurries to the line without substituting, the defense may be stuck with the eleven it has, the substitution freeze from In-Game Adjustments and Tempo. An offense that goes fast right after a big gain, or right after the defense has subbed into a run-stopping group, is usually hunting a mismatch it has already spotted.

Why second: personnel is visible before the formation is, and it's the context for everything the formation shows. As you'll see shortly, it adds surprisingly little to a run-or-pass call on its own. Its value is in what it sets up.

Step 3: formation, strength and the offensive tells

This is the richest step. Where is the quarterback: under center, in the shotgun, in the pistol? Where is the back, how deep, offset to which side? Where is the tight end, on the line or off it, in a three-point stance or standing up? How wide are the receivers? Which side is the formation strength? Part 2 taught the vocabulary, Parts 3 and 4 the families of plays each look goes with, and the tells catalogue below collects every alignment clue in one place.

Why third: the offense has to show its formation before the defense commits to its final picture, and the defense's last-second picture is a *response* to it. Read the formation first and you'll understand what the defense does next.

⁴Course calculation from nflverse play-by-play joined to FTN charting (quarterback alignment, tempo) and participation (personnel), 2025 regular season, runs and dropbacks with a down recorded. "Pass" is nflverse's *pass* flag, so sacks and scrambles count as passes, as in your log.

Step 4: motion response

When a player moves before the snap, watch the defense, not the man in motion. A defender who travels across the formation with him suggests man coverage; defenders who bump a spot over suggest zone; a safety who spins down turns two-high into one-high. That's the man/zone indicator from Motion and Shifts as Weapons. Then ask the offensive question: is he still moving when the ball is snapped (motion at the snap)? A jet motion at full speed is a run threat, or a decoy for one, to that side.

You will get to use this step on most snaps. PFF's charting had a shift or motion on 63.9% of 2025 snaps, the highest in its series, and every team but the Giants used one on at least half its plays.⁵

Why fourth: motion only happens after the offense sets, and its whole point is to make the defense answer a question before the snap. The answer is a clue for steps 5 to 7.

Step 5: the box

Count the defenders in the box and compare them with the blockers (box count). Six blockers against six defenders is a fair fight for a run; six against eight means someone will be unblocked, which is why an offense that sees eight usually throws or checks to a run away from the extra man (Run Fits and Box Math). A light box (six or fewer) against a shotgun offense is the classic invitation for a run or an RPO.

Why fifth: the box isn't final until motion has settled it. Count too early and you'll count a safety who is about to leave, or miss one who is about to arrive.

Step 6: the shell

One-high or two-high? Then the details: each safety's depth (13 yards is deep, 8 is down), his width (on the hash, outside it, in the middle), and any creep in the last seconds of the play clock. This was the spine of Part 6, and Disguise, Rotation, and Split-Field Coverage was about why the shell you see is often not the coverage you get. The league has moved toward showing two-high: NFL Next Gen Stats measured two-high coverage on 42.0% of dropbacks in 2025, the highest since it began tracking in 2018.⁶ Two-high before the snap, then, is no longer a tell by itself; the tell is which of the two safeties is shallower, narrower or moving.

Why sixth: safeties are the last defenders to settle and the most likely to be lying, so you read them as late as you can.

Step 7: leverage and depth

Look at the corners: pressed at the line or off at seven yards; on the receiver's inside shoulder or his outside. Leverage tells you what the defender is protecting and so where his help is: a corner playing outside leverage is giving up the inside because someone (a safety in the middle, a linebacker underneath) is there to take it. Then the linebackers' depth: at five or six yards they are playing the run and the short middle; creeping to three, they may be coming; backing out to eight before the snap on a passing down, they are getting a head start into zone drops.

Why seventh: leverage is the last thing a defense adjusts, and it only makes sense after you know the shell. An inside-leverage corner with two deep safeties means something different from the same corner with one.

Step 8: pressure tells

How many defenders are at or near the line, and could any of them drop? Linebackers mugging the A gaps, a safety or nickel back walked up beside the end, a defensive lineman sitting back on his heels: these are the pressure look tells from Part 7. Then the count that caps everything: **how many players are deep?** Two deep safeties and two off corners leave seven to rush and cover underneath, and a defense almost never rushes six with only one underneath. Finish with the offense's side: where is the back, which way will the line slide, where is the hot route?

Why eighth: pressure looks are deliberately last-second. Defenses walk up late so the offense has as little time as possible to change its protection, and you get the same late look the quarterback does.

⁵Daire Carragher, "NFL pre-snap motion usage reaches new highs across the NFL," PFF (Feb. 23, 2026): <https://www.pff.com/news/nfl-pre-snap-motion-usage-reaches-new-highs-across-the-nfl>. PFF's series counts shifts and motion together and is not comparable with FTN's `is_motion` flag, which nflverse defines as motion "before or at the time of the snap."

⁶NFL.com, "The coverage that turned the Seahawks into Super Bowl winners and is taking the NFL by storm" (Next Gen Stats): <https://www.nfl.com/news/the-coverage-that-turned-the-seahawks-into-super-bowl-winners-and-is-taking-the-nfl-by-storm> (two-high on 32.9% of dropbacks in 2018, 42.0% in 2025).

Step 9: call it, with a number

Say it out loud before the snap: run or pass, and how sure you are. Then, if you have a second left, one more layer: the coverage family, how many will rush, or where the ball should go. This is the step this chapter adds, and it gets its own section below. The rule is simple: the call is made *before* the snap, or it doesn't count.

What each step is worth

How much does each step actually help? For the run-or-pass call, we can measure it. Charting services record, for every NFL snap, much of what steps 1 to 5 ask about: the situation, the personnel, the quarterback's alignment and the number of backs, whether there was motion, whether the offense huddled, and how many defenders were in the box. (Steps 6 to 8 aren't in public data: nobody charts pre-snap safety depth or corner leverage for every snap.) So we can do what a reader would do: learn from one season what each combination of clues usually meant, then call every snap of the next season, adding one step at a time.

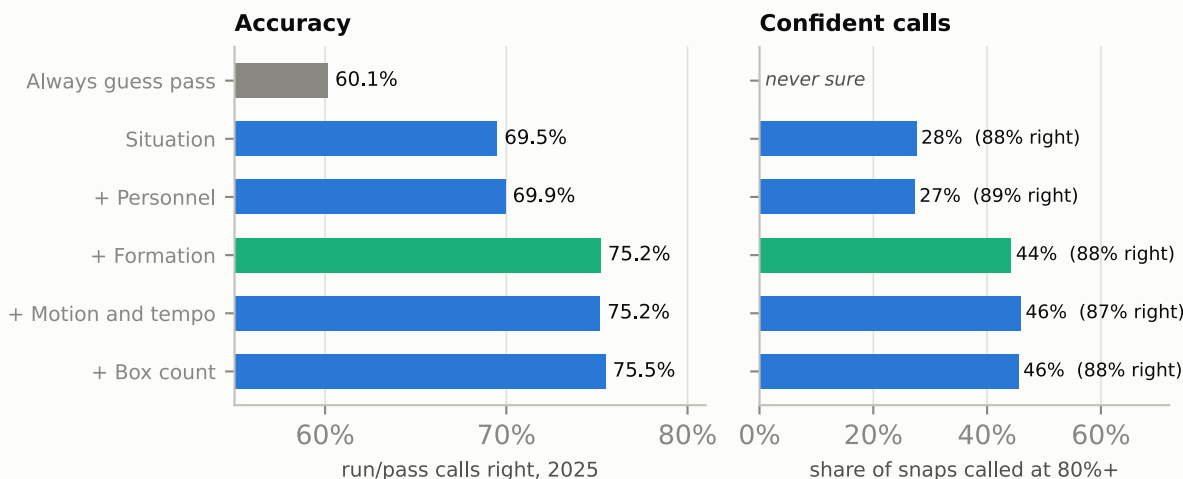


Figure 3: What the measurable steps are worth for a run-or-pass call. Each row adds one checklist step to a model learned on the 2024 regular season and then scored on every 2025 regular-season run and dropback (32,813 snaps). Left: how often the call was right. Right: the share of snaps the model could call at 80% confidence or more, and (in the label) how often those confident calls were right. The situation does most of the work; the formation (quarterback's alignment and number of backs) is the one big jump after it; motion, tempo and the box add almost nothing to run or pass. Source: nflverse play-by-play, FTN charting and participation.

Three lessons hide in that chart.⁷

The situation is most of the call. Down, distance, field zone, score and clock take you from 60% ("always pass") to 69%. That is the step a casual fan skips, because it feels too obvious to count. It isn't obvious; it's just quick.

The formation is the one big jump after it. Adding the quarterback's alignment and the number of backs lifts the model to 75%, and, as important, raises the share of snaps it can call at 80% confidence or better from 28% to 44%. Personnel alone added little on top of the situation (about 0.5 points), because most of what the personnel tells you, the formation tells you again, more precisely.

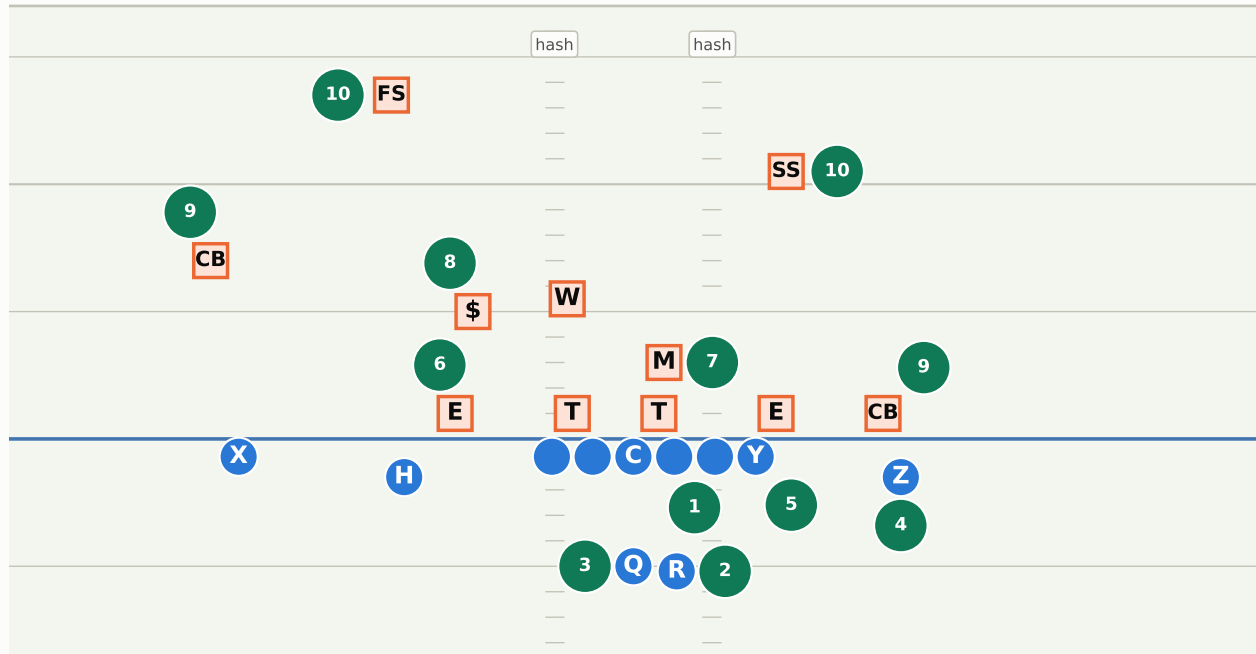
Motion, tempo and the box add almost nothing to run or pass. Together they move the needle by 0.3 points. That isn't because they're useless. It's because they answer *other questions*. Motion is for identifying the coverage. The box is for knowing whether the run, if it comes, has a chance. And the defense's box count is partly a reply to the offense's formation, so it repeats what you already knew. The same is surely true of the steps nobody charts: safety depth and corner leverage tell you little about *whether* the offense will throw, and a great deal about *where* and *against what*.

⁷Course calculation. Nested logistic regressions (ridge-penalised, fitted with numpy) on nflverse play-by-play, FTN charting and participation. Training set: 2024 regular-season runs and dropbacks; test set: all 32,813 2025 regular-season runs and dropbacks. Situation = down by distance (1, 2, 3, 4-6, 7-10, 11+ yards), field zone, score margin (five bands), last two minutes of a half, and fourth quarter by score; personnel = offensive personnel group from participation; formation = FTN qb_location by number of backs (n_offense_backfield, 0, 1, 2+); motion and tempo = FTN is_motion and is_no_huddle; box = participation defenders_in_box (5 or fewer, 6, 7, 8+). nflverse's own xpass model, which uses only the situation, calls 71% of the same snaps correctly.

That changes how you should spend your twenty seconds. **By the end of step 3 your run-or-pass call is essentially made. Steps 4 to 8 are for the second layer:** the coverage, the pressure, where the ball should go, which way the run will hit. A reader who spends all twenty seconds agonising over run or pass is using the expensive steps on the cheap question.

The tells catalogue

How to Watch a Broadcast started a catalogue with four starter tells (stance, the back's depth and offset, splits, safety depth) and promised the full version here. This is it: ten clues, five from each side of the ball, and a plate that shows where each one lives.



Offense

- 1 Linemen's stance**
Heavy hand, weight forward: run block. Light hand, hips back, a tackle's outside foot staggered back: pass set.
- 2 Back's depth and offset**
Shotgun runs usually go away from the side he's on; deeper than usual wants room to read a run, closer suggests a block or a quick route.
- 3 Shotgun, pistol, under center**
2025 pass rates: shotgun 79%, pistol 29%, under center 31% (and most of those were play-action).
- 4 Receiver split**
Reduced (inside the numbers): a crack block, a short crosser, or room to break outside. Wide: isolated, a deep or in-breaking route.
- 5 Tight end's stance**
Hand down, in line: usually a block (or a release that looks like one). Standing, flexed off the line: usually a route.

Defense

- 6 Lineman's width and stance**
A wide edge rusher is built to rush the passer; a lineman back on his heels may be stunting or dropping.
- 7 Linebacker depth**
At 5-6 yards: the run and the short middle. Creeping to 3 late: blitz or a run fit. Backing out before the snap: a zone drop.
- 8 Nickel's spot**
Apexed between the box and the slot: run support or a zone. Walked up and tight on the slot: man, or a nickel blitz.
- 9 Corner depth and leverage**
Pressed: man or a hard corner. Off and outside: help is inside. Inside leverage: help is over the top or outside.
- 10 Safety depth and width**
Deep and on or outside the hash: a deep half or quarter. Shallower, narrower or moving: the one about to rotate or come down.

Figure 4: The tells catalogue on one plate. A shotgun offense with 11 personnel (one back, one tight end) faces a nickel defense with two deep safeties. Numbered badges mark where each tell lives: 1 to 5 on the offense, 6 to 10 on the defense; the key below says what each usually suggests. Every one is a lean, not a guarantee, and some of the alignments drawn here are exaggerated so the clue is visible: the right corner is pressed with inside leverage, the left corner is off with outside leverage, the Mike has crept to three yards, and the right safety is shallower and narrower than his partner.

Reading the plate offense first, then defense, is deliberate: it is the order of the checklist. Offensive tells belong to step 3, when everyone is set; defensive tells to steps 6 to 8, when the defense has settled (or pretended to). Two of the stance tells are easier to understand from the side:

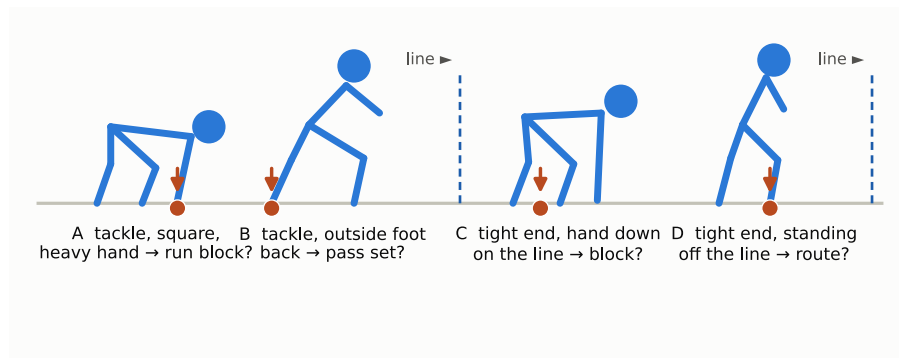


Figure 5: Two stance tells seen from the side (the line of scrimmage is to the right; the dot and arrow mark where the weight is). A and B: a tackle about to drive forward stands square, shoulders over a heavy hand; a tackle about to kick back into his pass set staggers his outside foot well back and sits on it. C and D: a tight end in a three-point stance on the line is set up to block (though a good one releases from the same stance); one standing up a yard off the line is set up to get into a route quickly. The tight end tell is weaker: many teams put the tight end in a two-point stance on every snap.

Here is the whole catalogue as a reference table: what each tell usually means, *why* it leaks (a tell that has no physical or tactical reason behind it won't survive a self-scout), how much to trust it, and whether you'll see it live or only on the replay.

#	Tell	Usually means	Why it leaks	How far to trust it; where you'll see it
1	Linemen's stance and weight	Heavy hand, weight forward: run. Light hand, hips back, a tackle's outside foot staggered: pass	A 320-pound man can't explode forward from his heels or kick back from his fingertips	Medium on early downs; quiet on short yardage, when everyone is heavy. <i>Replay (end zone); live only on close-ups.</i>
2	Back's depth and offset	Shotgun runs usually cross the quarterback's face, away from the back's side; a deeper back wants time to read	The handoff path points one way; reading blocks needs a running start	Medium for direction; weak for run or pass. <i>Live, every snap.</i>
3	Shotgun, pistol or under center	Shotgun mostly pass (79% in 2025); pistol and under center mostly run	Each alignment is built for a family of plays	Strong for run/pass, but under-center passes are mostly play-action, built to exploit this tell. <i>Live, every snap.</i>
4	Receiver split	Reduced: a crack block, a short crosser, or room to break outside. Wide: an isolated route, deep or breaking inside	Receivers set their split to make their job easier	Weak: tells you <i>where</i> to look more than <i>what</i> . <i>Live on the wide shot; replay otherwise.</i>
5	Tight end's stance	Hand down on the line: block likely. Standing, flexed: route likely	Blocking starts low; releasing starts tall	Weak; many teams stand the tight end up on every snap. <i>Live.</i>
6	Defensive lineman's width and stance	Wide-aligned edge: pass rush. A lineman sitting back: stunt or drop	A wide rusher has a straight path to the quarterback; a looper needs his weight back	Medium on obvious passing downs. <i>Replay; live from the end-zone shot.</i>

#	Tell	Usually means	Why it leaks	How far to trust it; where you'll see it
7	Linebacker depth	Creeping to 3 yards late: blitz or a downhill run fit. Backing out early: zone drop	A blitzer needs a head start; a dropper needs depth	Medium; walking up is also a bluff. <i>Live, on the wide shot.</i>
8	Nickel's spot	Apexed: run support or zone. Tight on the slot: man, or a nickel blitz	Man coverage needs proximity; an apex keeps both jobs in reach	Medium. <i>Live.</i>
9	Corner depth and leverage	Press: man or a hard corner. Off with outside leverage: help inside (a deep middle safety). Inside leverage: help outside or over the top	A defender takes away the side where he has no help	Medium; leverage also follows the play-call's coverage of the week. <i>Replay (skycam); live on the wide shot.</i>
10	Safety depth and width	Deep on the hash or outside it: a deep half or quarter. The shallower, narrower or moving one: the rotation	A safety can't cover more than about eight to ten yards of ground between the snap and the throw	Strong <i>when it moves late</i> ; disguise is the norm. <i>Top edge of the TV picture; skycam replay.</i>

A few of these deserve a sentence more than the table can give them.

The back's offset is a direction tell, not a run-or-pass tell. Shotgun runs most often send the back across the quarterback's face, because the mesh (the moment the ball is handed off) happens while the back is moving across. A back offset to the right usually runs left. But the same offset is where a back sets up to pass-protect against a rusher from that side, so it tells you *which* way a run goes far more than *whether* there will be one.

The split tell has two readings, and the book has given you both. A reduced split brings a receiver closer to the formation, which helps three different jobs: a crack block on a linebacker or safety inside him, a short crossing route (the trip across is shorter), and a route that breaks *outside* and needs room to get there. That's why reduced split is a weak tell for *what* is coming. Its real value is a *where*: when a receiver tightens his split on a run look, watch the defender inside him, because that's the man he's coming to block.

Leverage points at the help. A defender usually lines up to take away the side where he has no help, and funnels the receiver toward the side where he has some. So a corner with outside leverage and a single safety in the middle is telling you the safety is his help; a corner with inside leverage against two-high may have a safety over the top and outside. Read leverage as an arrow pointing at the help, and the coverage family usually follows.

Safety width is the most underused tell. Everyone counts deep safeties; few look at how far apart they are. A two-high safety standing on the hash or outside it has a deep half or deep quarter to cover and can't reach the deep middle in time. A two-high safety cheated inside the hash, eight yards or fewer from the middle, *can*, and the one standing shallower than his partner can come down. That asymmetry, not the two-high picture itself, is how you see a rotation coming.

Common misconception

"NFL players are too good to have tells." Every player has tendencies, because the best way to do a job quickly is to start in the position the job needs. What separates good teams from bad ones isn't the absence of tells, it's what they do about them. Every staff self-scouts its own film looking for the clue an opponent could read, and then decides whether to break it, hide it, or keep it as bait (Constraint Theory and Sequencing). The Rams' under-center passing in 2025 is the textbook bait: the alignment says run, so the linebackers step up, and the play-action pass goes behind them. A tell that is right 70% of the time is still worth using. You just shouldn't be 95% sure of it.

Stated confidence: putting a number on it

Step 9 asks for a number, and this is where the course's most important habit gets its name. A **stated confidence** is the probability you attach to a prediction before the snap, said or written as a number rather than a feeling: not “probably a pass” but “pass, 75%.” The prediction log has had a confidence column since Part 1. From here on, it is not optional.

Why a number, rather than just the call?

1. **It separates a read from a guess.** A 55% call and a 90% call that both miss are different mistakes. The first was honest uncertainty; the second was a misread. Without the number, your log can't tell them apart.
2. **It forces you to use the base rate.** You can't say “80%” for a pass on first-and-10 without asking what moved you from 50. If nothing did, the honest number is 50 to 55.
3. **It can be scored.** A forecaster who says “70% chance of rain” isn't wrong when it stays dry; she's wrong if, across all her 70% days, it rains on nine in ten, or on three in ten. Your calls work the same way: a 70% call should come true about seven times in ten. The next chapter, Film Study, scores the season's log that way, with a Brier score (the average squared gap between your stated confidence and what happened) and a calibration plot (do your 70% calls come true about 70% of the time?). Researchers who study forecasters have found that saying the number, rather than a word like “probably”, is itself one of the habits that makes forecasters better.⁸

How do you choose the number? Start from the base rate (step 1), move it with the strong clues (the formation is the big one), and then respect two limits:

- **Never below 50.** Your call is whichever side you think is more likely, so its confidence is at least 50%. A 50% call is perfectly honest: it says “I have nothing”, and logging it is better than skipping the snap.
- **Rarely above 90.** Offenses choose when to break their tendencies, and the snaps that break them are the ones built to fool you. Even the most lopsided situations in football leave room for a surprise: third-and-9 is a pass 94% of the time, not 100%. A reader who logs 99% will be badly wrong once a quarter and will learn nothing from it.

What does a well-judged spread of confidences look like? Here is the model from the previous section (steps 1 to 3 only: situation, personnel and formation) on every 2025 snap, grouped by how sure it was:

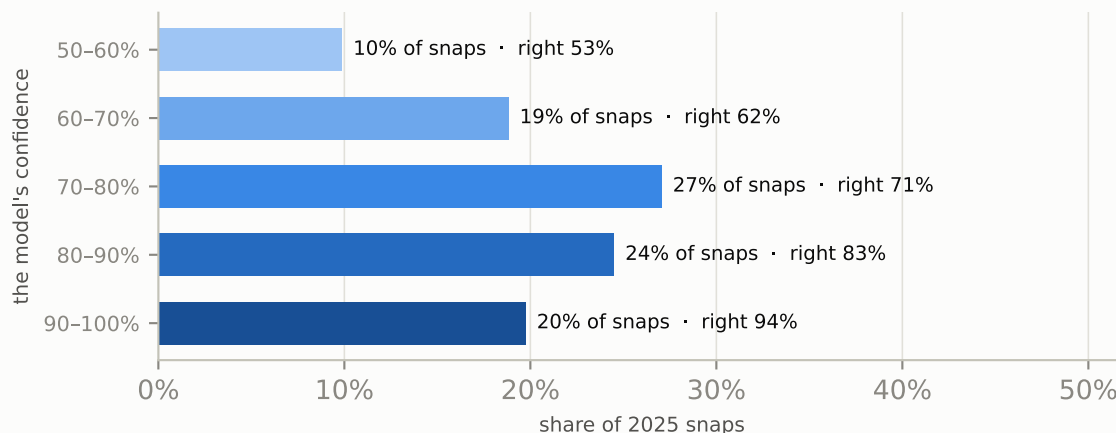


Figure 6: What honest confidence looks like for run or pass. Every 2025 regular-season run and dropback, grouped by how confident a model using only steps 1 to 3 (situation, personnel, formation; learned on 2024) was in its call. Bars: the share of snaps in each band. Labels: how often the call in that band was right. The calls come true at close to the rate they claim (two to four points less often, because the model learned on a different season), and about one snap in five really is a 90%-plus call (long-yardage downs, empty backfields, the end of halves). The rest are spread across the middle, where most of football lives. Source: nflverse play-by-play, FTN charting and participation.

That chart is the target for your own log. When the model says 70 to 80%, it's right 71% of the time; when it says 90% or more, it's right 94% of the time. Nobody expects a person to match a model learned on 33,000 snaps, but the *shape* is the goal: confident where the situation is lopsided, modest everywhere else, and right about as often as you claimed.⁹

⁸Philip E. Tetlock and Dan Gardner, *Superforecasting: The Art and Science of Prediction* (Crown, 2015), on the forecasting tournaments in which forecasters who used fine-grained numerical probabilities, kept score and updated often were more accurate.

⁹Same model and data as the chart of what each step is worth, using steps 1 to 3 only. Confidence is the model's probability for whichever side it called.

A small vocabulary helps you say the number fast:

Say	Means	Typical use
“Coin flip”	50%	Nothing to go on: the formation was off screen, or the clues pull both ways
“Lean”	55–60%	First-and-10 from a balanced look; one weak clue
“Probably”	65–70%	The formation agrees with the situation
“Likely”	75–80%	A strong formation clue and a situation that points the same way
“Near certain”	85–90%	Third-and-long, an empty backfield, a two-minute drill when trailing

Layered predictions

Run or pass is the first layer, and after step 3 it’s mostly decided. A good reader makes a second call on the snaps where there’s time, and states a confidence for it too:

- **On a likely pass:** the coverage family (man or zone; one-high or two-high after the snap), the rush (four, or five or more), or where the ball should go (the quick throw, the sticks, a shot).
- **On a likely run:** the direction (to the strength, away from the back’s offset, at the light side of the box) or the scheme family (zone, if the line steps together; gap, if a guard is set to pull).

Second-layer calls are harder, and your confidence should show it. The base rates help here too. In 2025, defenses sent five or more rushers on only 27% of dropbacks (39% on third-and-7 to 10), so “four rush” is the default until a pressure tell moves you off it. Zone was the coverage on about 69% of charted 2025 dropbacks, so “zone” is the default coverage call; FTN’s man-coverage share has jumped around from season to season in a way that looks like a change in charting rather than in football, so treat that number as rough.¹⁰ A second-layer call at 60–70% is a good call. Few are better than that, and anyone who claims to call coverages at 90% before the snap is either watching a very predictable defense or not keeping score.

Madden vs. real life

In Madden you pick a play from a menu, and when you’re on defense you see the offense’s formation and pick a coverage, all with perfect knowledge of your own call. Watching a real game is the reverse: you know neither call, and neither does anyone on the field. The quarterback’s pre-snap read *is* this checklist, run by the person with the most at stake, and he gets it wrong often enough that offenses build hot routes and post-snap reads for the times he does. So a probability isn’t a hedge. It is the most accurate description of what anyone, including the players, knows before the snap.

Common misconception

“My 90% call missed, so I was wrong.” Not necessarily: one in ten 90% calls *should* miss. You can’t judge a single probabilistic call by its outcome, only a pile of them. That’s why the log matters, and why Film Study scores a season of calls rather than a play. The mistakes to look for are patterns: 90% calls that miss one time in three (overconfidence), or a log full of 55% calls that come true 75% of the time (you know more than you’re admitting).

¹⁰Course calculation from nflverse participation (`defense_man_zone_type`) and FTN charting (`n_pass_rushers`), 2025 regular-season dropbacks with a rush count recorded; coverage share among those charted man or zone. On the season-to-season jumps in FTN’s man share (28.8% in 2022 under the earlier NGS source, then 41.2%, 48.3% and 30.8% for 2023–2025), see the course’s notes on the 2022/2023 source break in Tendencies.

Film room: one drive, nine snaps

i Film room: Seahawks 2025 — the NFC Championship Game

The game: Rams at Seahawks, NFC Championship Game, January 25, 2026, at Lumen Field. Seattle, the NFC's top seed (14–3), won 31–27 and two weeks later beat New England 29–13 in Super Bowl LX.¹¹ In 2025 Seattle's head coach was Mike Macdonald, its offensive coordinator Klint Kubiak (whose system comes from the Shanahan family tree; see Shanahan's Wide Zone Tree) and its quarterback Sam Darnold.¹²

The drive: Seattle had gone up 24–13 early in the third quarter; the Rams answered with a touchdown to make it 24–20 with 9:45 left in the quarter. After the touchback, Seattle drove 65 yards in nine snaps and scored on third-and-3 from the Rams' 13, a 13-yard pass to Cooper Kupp, to go up 31–20 with 4:52 left in the third.

Why this drive: it has a bit of everything the checklist is for: three different personnel groups, under center and shotgun, an empty backfield, a screen against a blitz, play-action, a no-huddle snap, two third downs and a red-zone touchdown, against a defense that played base, nickel and dime and three different coverage labels.

The sources: every pre-snap fact below (personnel, the quarterback's alignment, the number of backs, motion, tempo, box count, defensive personnel) and every post-snap one (rushers, coverage label, time to throw) comes from the public charting in nflverse: the play-by-play, the participation file and FTN's charting.¹³ The diagrams are typical pictures of what was charted, not the film: the charting doesn't record exact alignments or the routes of receivers who weren't thrown to.

Here is the drive as your log would record it. The calls weren't made by watching the game. They were made by rule from the pre-snap columns only, so hindsight can't creep in: the run-or-pass call and its confidence come from the step 1 to 5 model, fitted on the 2024 and 2025 regular seasons; the coverage and rush calls are the 2025 base rates for that down, distance, field zone and number of defensive backs.

¹¹Score, date and site from nflverse schedules and play-by-play (game 2025_21_LA_SEA: Seattle 31, Los Angeles Rams 27, January 25, 2026, Lumen Field); also "2025–26 NFL playoffs," Wikipedia: https://en.wikipedia.org/wiki/2025%E2%80%9326_NFL_playoffs . Seattle's 14–3 record and NFC top seed, and Super Bowl LX (Seattle 29, New England 13, February 8, 2026): https://en.wikipedia.org/wiki/Super_Bowl_LX .

¹²Seattle's 2025 staff (Macdonald head coach, Kubiak offensive coordinator, Aden Durde defensive coordinator) as listed for Super Bowl LX: https://en.wikipedia.org/wiki/2026_Seattle_Seahawks_season and https://en.wikipedia.org/wiki/Super_Bowl_LX . For 2026, Kubiak became the Raiders' head coach and Brian Fleury is Seattle's offensive coordinator. Seattle signed Sam Darnold on March 13, 2025, and Cooper Kupp, released by the Rams, on March 14, 2025 (three years, about \$45 million): <https://www.si.com/nfl/cooper-kupp-signs-seahawks-rams-release> ; both names appear on Seattle's snaps in the game's play-by-play.

¹³nflverse play-by-play (desc, down, distance, yard line, air_yards, yards_after_catch), participation (offense_personnel, defense_personnel, defenders_in_box, defense_coverage_type, defense_man_zone_type, time_to_throw, was_pressure, route) and FTN charting (qb_location, n_offense_backfield, is_motion, is_no_huddle, is_play_action, is_screen_pass, n_pass_rushers, n_blitzers, read_thrown, is_qb_out_of_pocket), game 2025_21_LA_SEA, Seattle's drive beginning at 9:41 of the third quarter. Charting is done by people and contains errors; the labels are FTN's, not the teams'.

PREDICTION LOG • NFC Championship, Rams at Seahawks • Seattle offense, Q3								
#	Situation	Offense shows	Defense	Call	If pass	What happened	R/P	Pass layer
1	1st & 10 SEA 35	12 · UC · 1 back motion	base box 7	Run 55%	zone 80% 4 rush 65%	Holani run, +3	✓	-
2	2nd & 7 SEA 38	12 · UC · 2 backs motion	nickel box 6	Run 60%	zone 75% 4 rush 80%	Walker run, -2	✓	-
3	3rd & 9 SEA 36	11 · gun · 1 back	dime box 5	Pass 90%	zone 65% 4 rush 75%	Kupp catch, +12 (1st)	✓	✓ ✓
4	1st & 10 SEA 48	13 · UC · 2 backs motion	base box 7	Run 70%	zone 80% 4 rush 65%	Walker run, +11 (1st)	✓	-
5	1st & 10 LA 41	11 · gun · empty	nickel box 6	Pass 85%	zone 80% 4 rush 80%	Holani catch, +9	✓	✓ ✓
6	2nd & 1 LA 32	11 · gun · 1 back motion	nickel box 6	Run 55%	zone 70% 4 rush 75%	Smith-Njigba screen, +12 (1st)	✗	✓ ✗
7	1st & 10 LA 20	12 · UC · 2 backs motion	base box 7	Run 75%	zone 65% 4 rush 55%	Play-action, Barner +7	✗	✓ ✓
8	2nd & 3 LA 13	12 · UC · 1 back motion, no-huddle	base box 7	Run 70%	man 70% 5+ rush 80%	Incomplete to Bobo	✗	✗ ✓
9	3rd & 3 LA 13	11 · gun · 1 back motion	nickel box 6	Pass 75%	man 55% 4 rush 75%	Kupp, 13-yard TD	✓	✗ ✓
Run/pass: 6 of 9 right. Pass layer on the 6 dropbacks: coverage 4 of 6, rushers 5 of 6.								

Figure 7: The drive as a prediction log: Seattle's nine snaps from its own 35 to the end zone, NFC Championship Game, third quarter. Situation, offense and defense are the pre-snap facts in the charting (UC = under center; M = motion; NH = no-huddle; box = defenders in the box). The calls were made by rule from those facts alone: run or pass from the checklist model, the 'if pass' layer (coverage family, rushers) from 2025 base rates for the situation and the number of defensive backs. The last two columns score the calls; the pass layer is scored only on dropbacks, because coverage isn't charted on runs. Source: nflverse play-by-play, participation and FTN charting, game 2025_21_LA_SEA.

And here is how each call was built, step by step, the way the confidence should move in your head:

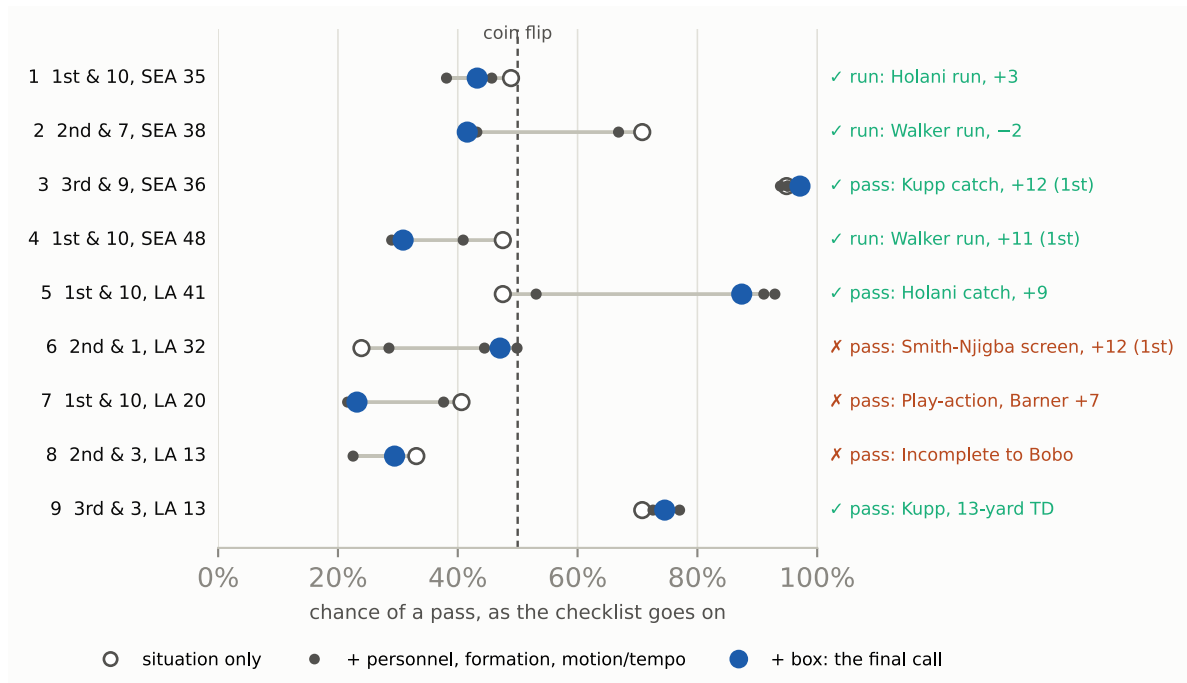


Figure 8: How the run-or-pass confidence moved through the checklist on each snap of the drive. Each row is one snap. The hollow circle is the probability of a pass from the situation alone; the small dots add personnel, then the formation (quarterback alignment and number of backs), then motion and tempo; the filled marker adds the box count and is the final pre-snap call. The right-hand label is what happened, ticked or crossed against the call. The biggest moves come at the formation step, and the misses are the second-and-1 screen (a coin flip that landed the other way) and snaps 7 and 8, passes from under center with two tight ends, exactly the look offenses use to break the tendency. Model fitted on the 2024 and 2025 regular seasons; source: nflverse play-by-play, participation and FTN charting.

And the drive in pictures, one pre-snap look per snap:

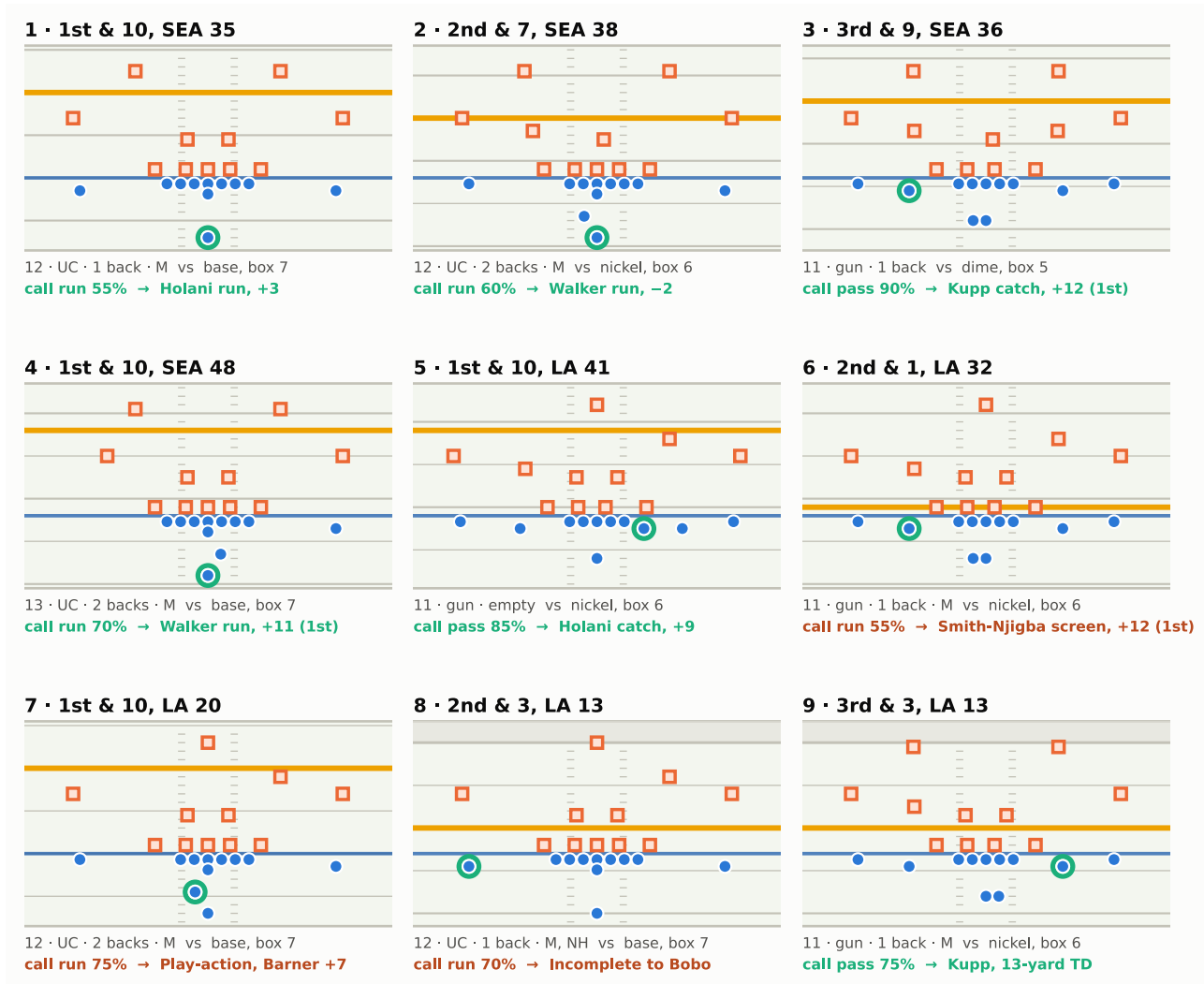


Figure 9: The drive in pictures: a typical pre-snap look for each of the nine snaps, built from what was charted (offensive personnel, quarterback alignment, number of backs, defensive personnel and box count, all exact), with the ball on the right yard line and the yellow line to gain. Everything else is illustrative: the charting doesn't record exact alignments, and for runs the safeties are drawn two-high because the coverage isn't charted on runs. For passes the shell is drawn to match the charted coverage family.

The ringed player is the ball carrier or the receiver the ball went to. Offense: blue circles; defense: orange squares.

Now walk it the way you would live, step by step. The snap numbers match the log.

Snap 1: first-and-10 at the Seattle 35. *Situation:* a four-point lead in the third quarter, a coin-flip down. *Personnel:* 12 (two tight ends); the Rams stay in base. *Formation:* under center, one back, with motion. Under center with two tight ends on first down is a run look, and the call is run, 55%, only a lean, because Seattle throws from this look too. *Box:* seven defenders against seven blockers (five linemen, two tight ends), a fair fight. George Holani runs behind the right guard for three. ✓

Snap 2: second-and-7. Second-and-medium leans pass, but Seattle stays in 12 personnel and puts a second player in the backfield (with one running back and two tight ends on the field, most likely a tight end as an H-back): two backs, under center. That moves the call back to run, 60%. The Rams answer with a nickel and a six-man box, which is the defense betting on a pass, and Kenneth Walker loses two yards running around the right end. The call was right; the run wasn't. That distinction matters for your log: you're scoring your read, not the offense.

Snap 3: third-and-9 at the 36. Here the run-or-pass question answers itself (pass, 90%), and everything after step 3 goes into the second layer. Seattle goes to 11 personnel in the shotgun; the Rams go to dime (six defensive backs) with only five men in the box. Six defensive backs and four rushers is a defense that wants to keep everything in front of it and tackle short of the sticks. The base-rate calls say zone and a four-man rush, and that's what came: four rushers, and a zone FTN labelled Cover 9 (a number that means different things to different staffs; see Coverage Fundamentals). Darnold held the ball 2.8 seconds and found Kupp short, a throw FTN charted as a checkdown: six yards in the air and six after the catch, a first down by three yards.

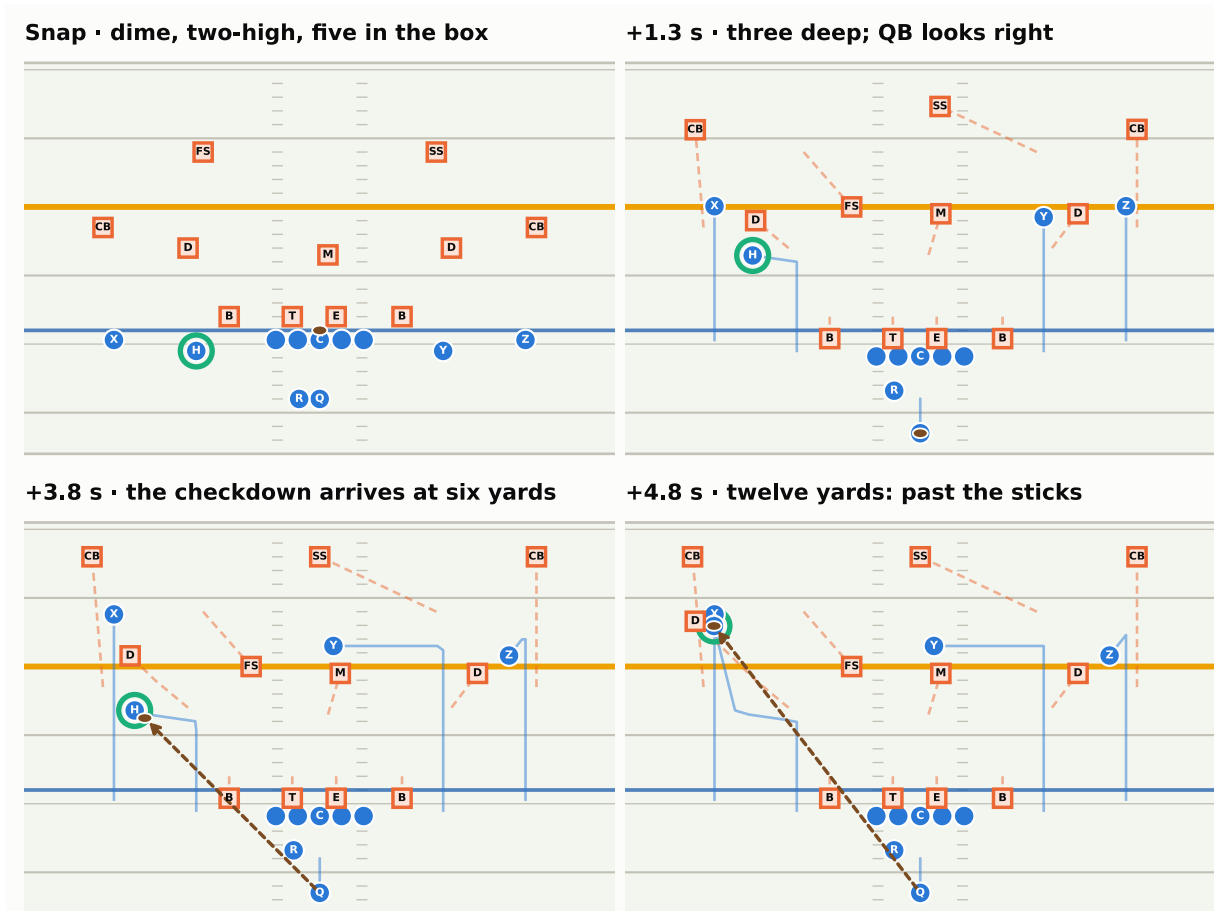


Figure 10: Snap 3, third-and-9 (a reconstruction, simulated in Big Data Bowl format). Charted facts: 11 personnel in the shotgun, no motion; the Rams in dime with five in the box; four rushers; a zone labelled Cover 9; a throw after 2.8 seconds to Cooper Kupp (H, ringed) running a quick out on the left, six yards in the air, with six more after the catch for the first down. Illustrative: the alignments, the other routes, and the rotation, drawn as one common meaning of ‘Cover 9’ (two-high before the snap, the safety away from the tight end coming down, three deep). What to notice: the underneath defenders sink toward the sticks and the quarterback looks right first; the conversion comes from the yards after the catch.

The lesson of snap 3 is the most common lesson of third-and-long. The run-or-pass call is free; the defense’s plan is easy to read from its personnel (six defensive backs, five in the box: keep it in front, rally, tackle); and the play is decided by the thing you can’t predict, a receiver breaking a tackle or not. Your second-layer call (“zone, four rush, the ball goes short and the game is the tackle”) was right, and that’s as good as reading gets.

Snap 4: first-and-10 at the 48. Seattle goes heavy: 13 personnel (three tight ends, one receiver), under center, two backs, motion. The Rams match with base and *two* nose tackles, a seven-man box. The call is run, 70%, as confident as a first-down run call gets. Walker runs behind the right guard for eleven. A heavy formation tells you *what*; it doesn’t tell you whether it will work.

Snap 5: first-and-10 at the Rams’ 41. The reverse picture: 11 personnel, shotgun, and **nobody in the backfield**. An empty backfield is the strongest pass tell in the catalogue (95% of 2025 empty-backfield snaps were dropbacks), and the call is pass, 85%. The second layer is the interesting part. Empty means five receivers and only five blockers, so a defense that brings a sixth man gets a free rusher; that’s why empty is so often a quick-throw formation. The Rams rushed four and played Cover 3, and Darnold checked the ball down to Holani, the running back, who had lined up outside the backfield as a receiver, on a shallow crossing route for nine.

Snap 6: second-and-1 at the 32. The most interesting snap of the drive. Second-and-short is the down where offenses take chances, because even a failed play leaves a manageable third down; from the shotgun with 11 personnel it was a pass 57% of the time in 2025. The full model, with the motion and the six-man box added, makes it a coin flip the other way: run, 55%. The second layer said zone (70%) and a four-man rush (75%). The Rams played zone but sent five, a blitz, and Darnold got the ball out in 0.7 seconds on a screen to Jaxon Smith-Njigba, who caught it four yards *behind* the line and ran sixteen yards after the catch. **X** on run or pass, and a 55% call is allowed to miss: that is what 55% means.

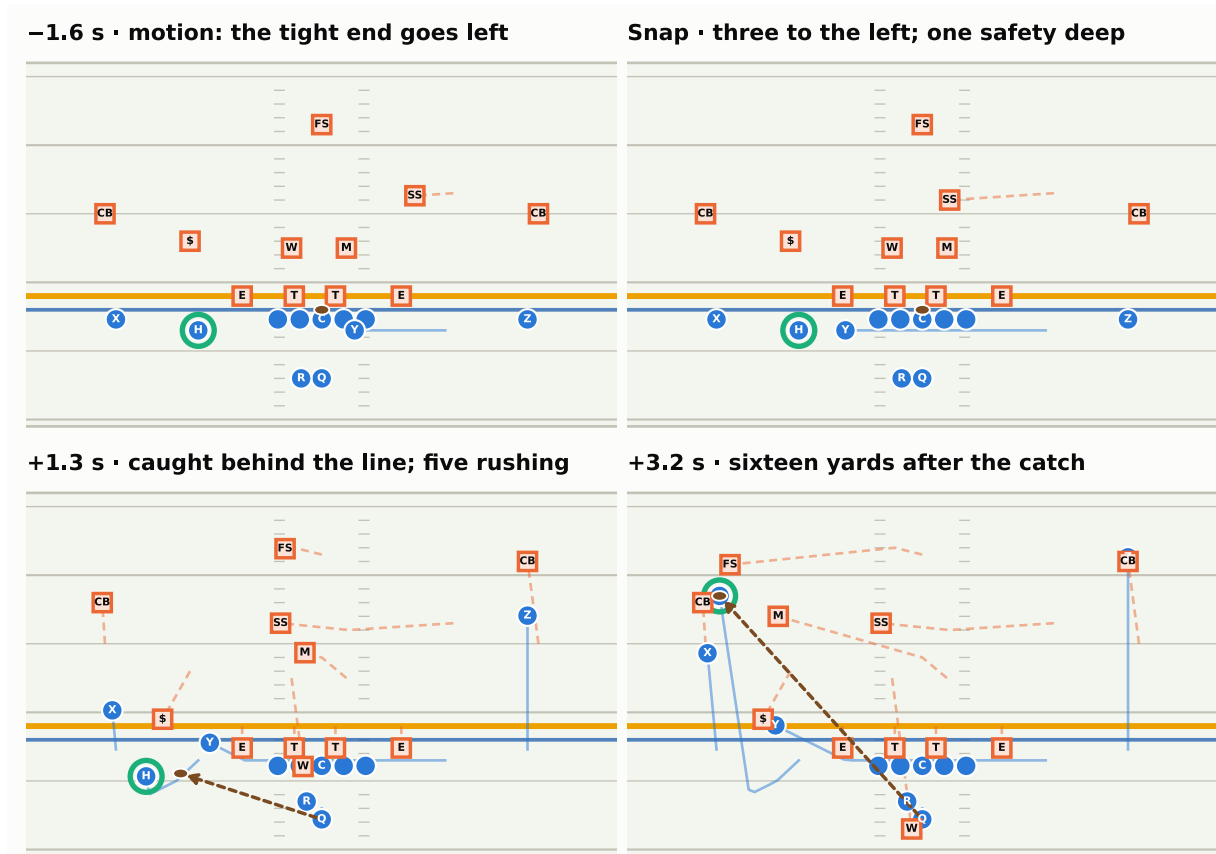


Figure 11: Snap 6, second-and-1 (a reconstruction, simulated in Big Data Bowl format). Charted facts: 11 personnel in the shotgun with motion; the Rams in nickel with six in the box; five rushers (one of them a blitz) and Cover 3 behind them; a screen to Jaxon Smith-Njigba (H, ringed) on the left, thrown 0.7 seconds after the snap, caught four yards behind the line, sixteen yards after the catch. Illustrative: who motioned (here the tight end, Y, across to the left), which defender blitzed (here the Will linebacker), and the blocks. What to notice: the fifth rusher is a defender who isn't in coverage on the screen side, and the ball is out before he arrives.

Why does a screen beat a blitz? Because every extra rusher is a defender who isn't covering anyone, and a screen deliberately lets the rushers through to put the ball where they came from. A pressure call against a screen is the worst outcome for a defense, and it happens because the defense was guessing too: second-and-1 from the shotgun is a coin flip for everyone. This is the snap where step 8 earns its place. If you'd seen a linebacker creeping toward the line and counted only one safety deep, your second-layer call would have moved from "four rush" toward "five or more, ball out fast", and the base rate would have been wrong for a reason you could see.

Snap 7: first-and-10 at the Rams' 20. Back to 12 personnel, under center, two backs, motion, against base and a seven-man box. Every step says run: run, 75%. It was play-action. Darnold faked the handoff, left the pocket and, after 4.7 seconds, found the tight end AJ Barner short on the left for seven. **X**. This is the miss you *should* have, the tendency-breaker built to beat the reader who believes the look: an offense from the Shanahan tree lives on making its run and play-action looks identical (Play-Action, Bootlegs and Screens). At 75% the call was honest. At 95% it would have been a misread.

Snap 8: second-and-3 at the 13, no huddle. Seattle hurried to the line in the same personnel, under center with one back. The call is run, 70%, and it missed again: a quick throw to Jake Bobo, out 0.6 seconds after the snap, incomplete. The second layer split: in this spot (second-and-short in the red zone, four defensive backs) the 2025 base rates said man (70%) and five or more rushers (80%). The Rams did send five, but played Cover 3 behind them. Two run-call misses in a row, both passes from a run look, is not bad reading; it's a play-caller deliberately working against his own tendency in the red zone. The between-snaps habit (line 13 on the card) is where you'd note it: *12 personnel under center, inside the 20, two passes in a row*. If they come back to it, your next call from that look should drop toward 55%.

Snap 9: third-and-3 at the 13. Shotgun, 11 personnel, motion: the model says pass, 75%. Third-and-short in the red zone is one of the few spots where man coverage is about as common as zone (the 2025 base rate for this spot was man 55%), so the coverage call was close to a coin flip. The Rams played zone, a Cover 6 (quarters to one side, Cover 2 to the other), and rushed four. They

got pressure, but Darnold threw on time, 2.4 seconds after the snap, to a receiver in his progression rather than a checkdown:¹⁴ Kupp on a short post, caught at the 6, six more yards after the catch, touchdown. ✓

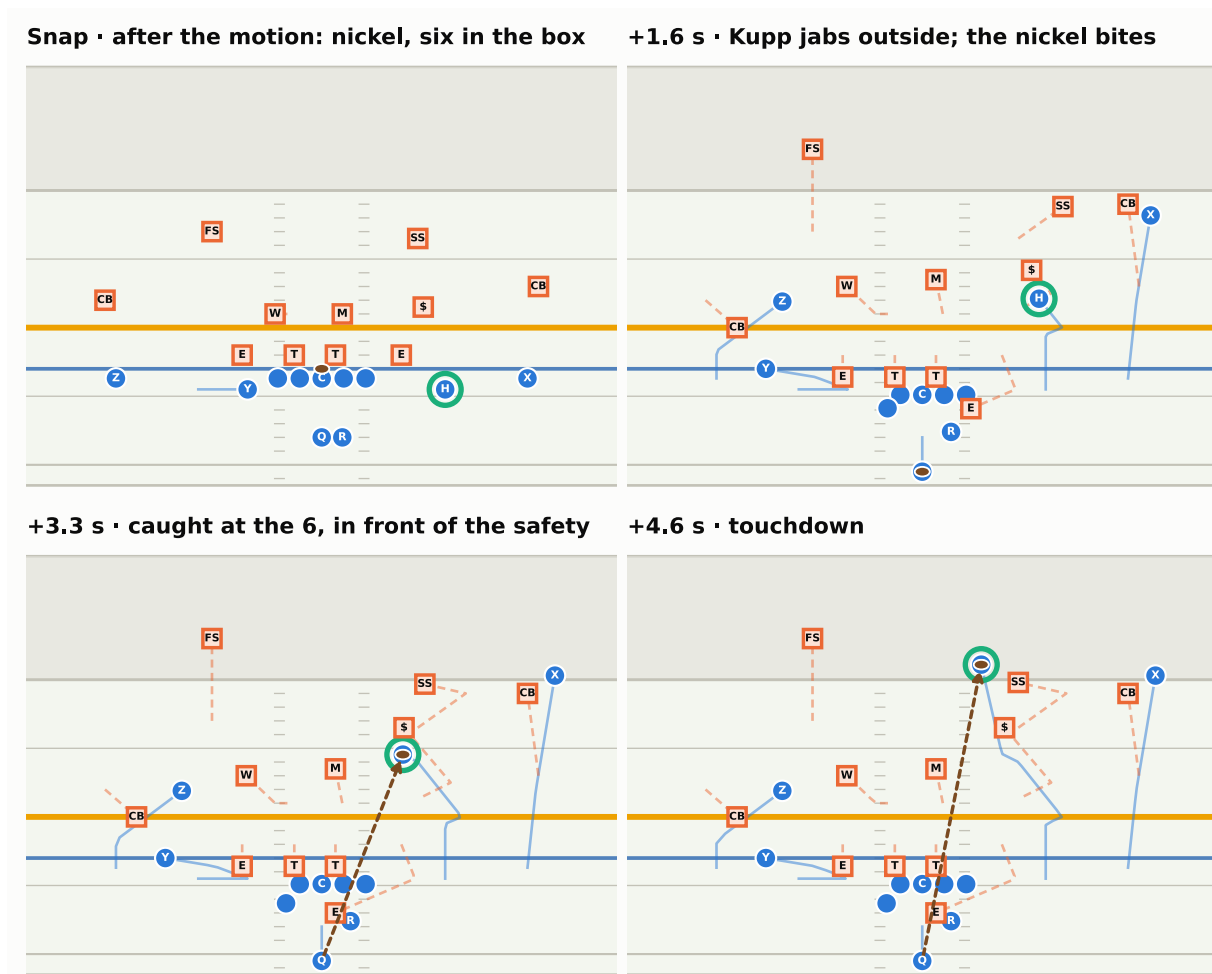


Figure 12: Snap 9, third-and-3 at the Rams' 13 (a reconstruction, simulated in Big Data Bowl format). Charted facts: 11 personnel in the shotgun with motion; nickel, six in the box; four rushers, with pressure; Cover 6; the ball thrown inside the progression (not a checkdown) to Cooper Kupp (H, ringed), on a post from the right side, seven yards in the air and six after the catch for the touchdown. Illustrative: the alignments, the tight end's motion (Y), the other routes, which rusher got home, and the coverage drawn as quarters on Kupp's side and Cover 2 on the other. Here the quarters safety on Kupp's side widens to help on the fade outside him and comes back too late. What to notice: in the red zone the deep defenders are already near the goal line, so a short post that gets inside the nickel and in front of the safety has room to run.

Add it up. The run-or-pass calls went 6 for 9. The three misses were all passes from looks that lean run: a coin flip on second-and-1, then play-action and a quick throw from under center with two tight ends, the looks an offense uses to spend its tendency. On the 6 dropbacks the second layer went 4 for 6 on the coverage family and 5 for 6 on the rush count. The coverage misses both came in the red zone, where the base rate leaned man and the Rams played zone; the rush miss was the second-and-1 blitz. That's a realistic live result for a good reader, and it says where the extra points are: in the red zone and on second-and-short, steps 4, 6 and 8 (the motion response, the shell and the pressure tells) are worth more than the base rate, so look harder there before you trust it.

Three habits from the drive are worth keeping:

¹⁴FTN's `read_thrown` field records which read in the quarterback's progression the ball went to, with "CHK" for a checkdown, "DES" for a designed throw with no progression (screens, many RPOs) and "SD" for a scramble drill; snaps 3, 5 and 7 of this drive are coded CHK, snaps 6 and 8 DES. This touchdown is coded "1". The nflverse data dictionary (https://nflreadr.nflverse.com/articles/dictionary_ftn_charting.html) says "0" is the first read and "1" the second, but in the 2025 file "0" appears on almost every run and on only about 2% of passes with air yards recorded, while "1" is by far the most common code on those passes (over half), which suggests "1" is in practice the primary read. Either way, it was a read in the progression, not a checkdown or a designed throw.

- **Score the read, not the play.** Snap 2 was a correct run call on a run that lost yards. Snap 3 was a correct pass call on a checkdown that only converted because of the yards after the catch. Your log grades you on what you could know before the snap.
- **Expect the tendency-breaker where it is cheapest.** Play-action from a heavy look on first down, and a quick pass from the same look in the red zone, are the classic places an offense spends a tendency. A 65% call that misses there is fine; a 90% call that misses there is a lesson.
- **Notice the defense's bets.** Dime with five in the box on third-and-9, a nickel against two tight ends on second-and-7, five rushers on second-and-1: each is the defense making its own prediction, and each is a clue to the second layer.

i Go deeper: the same drive in real tracking data

The figures above are simulations in the Big Data Bowl's tracking format, built from the charting. The NFL releases real player-tracking data only for the seasons and weeks of each year's Big Data Bowl, and no public release covers the 2025 playoffs (the 2026 edition used the 2023 regular season). For a play that is in a release, the real version is one call to `gridiron.bdb.prepare()` away (the code is in the web version of this page), and Tracking Data and the Big Data Bowl walks through it.

Between snaps: keeping the read up to date

The checklist is per-snap, but the best clue about the next snap is often the last twenty. Two habits keep your read current, and both happen while the broadcast is doing something else.

After every snap (five seconds, during the replay): write the log line. The call, the confidence, the clue that decided it, the result. That's line 12 on the card, and it's the habit How to Watch a Broadcast started. Don't skip the clue column: after a few games it tells you which tells are working *for you*.

At every stoppage (a TV timeout, the end of a quarter, a change of possession): update the picture of this team, today. Three questions:

1. **What are they doing from each look?** "Three runs from 12 personnel under center on first down" is a game tendency. It is more current than any season number, and play-callers know it too, so expect the tendency breaker when the streak gets long. That's the honest version of the "they're due" instinct: not that a pass is owed, but that the play-caller is watching the same tally you are.
2. **What did they just change?** Staffs adjust between series, not just at halftime: a new protection against a rusher who is winning, a coverage rolled toward a receiver who is hurting them, a run that worked coming back with a twist (In-Game Adjustments and Tempo). The call after a successful play is often built on it.
3. **What does the clock want?** The score and the time left change the base rate before step 1 even starts: a team ahead late runs, a team behind throws, and inside the last two minutes of a half the clock rules from The Clock, the Two-Minute Drill, and Fourth Down take over. Fourth down is its own prediction: go, kick or punt, before the offense decides.

When the offense goes fast

Against tempo the twenty seconds shrink to five or ten, and the full checklist won't fit. Run the short one: **1, 3, 6, 9**. Situation (you already have it from the last snap, plus the new down and distance), formation (the single biggest clue after the situation), shell (the single most useful second-layer clue), and the call. Drop personnel (it usually hasn't changed; that's the point of tempo), motion (tempo offenses use less of it), and the box and leverage counts you can't finish anyway. The measured value of what you drop is small for run or pass; the cost is mostly in your second layer, so lower those confidences when the offense is hurrying.

The same applies when the broadcast is still on a replay at the snap, which happens more than it should. Call it from the situation alone, write "sit" as the clue, and move on. A 60% call from the scorebug is still a call.

The log, final form

Your log has grown a column at a time since Part 1. Here is its final form, the one Film Study will score. Keep the columns from How to Watch a Broadcast (snap, quarter, clock, down, to-go, ball on, score) and add or tighten these:

Column	What you write	Example
<code>pers</code>	Offensive personnel, two digits	<code>12</code>
<code>align</code>	QB alignment and number of backs	<code>U1</code> (under center, one back), <code>S0</code> (empty)

Column	What you write	Example
motion	Motion or shift before the snap	M or blank
box	Defenders in the box, if you counted	7
shell	One-high or two-high, pre-snap	1 or 2
call	Run or pass	R
conf	Your stated confidence, as a decimal	0.65
clue	The one thing that decided it	align
l2	Your second-layer call	zone, 5+, left
l2conf	Its confidence	0.6
result, l2result	What happened, filled in after the snap	P, zone

At halftime, a few lines of pandas tell you how the half went. Here it is run on the drive above, saved as a CSV the way you'd type it:

```
import pandas as pd

log = pd.read_csv("nfc_title_q3.csv")
log["hit"] = log["call"] == log["result"]
print(f"Right on {log.hit.sum()} of {len(log)} calls ({log.hit.mean():.0%}); average stated confidence {log.conf.mean():.0%}")
bands = pd.cut(log.conf, [0.5, 0.6, 0.7, 0.8, 0.91], right=False, labels=["50-59", "60-69", "70-79", "80-90"])
print(log.groupby(bands, observed=True).hit.agg(calls="size", right="mean").round(2).to_string())
```

```
Right on 6 of 9 calls (67%); average stated confidence 71%
      calls  right
conf
50-59      2    0.5
60-69      1    1.0
70-79      4    0.5
80-90      2    1.0
```

Two numbers to watch over a season, not a half: your hit rate against your average stated confidence (if you're right 70% of the time while claiming 80%, you're overconfident), and the hit rate within each band (your 80% calls should come true about 80% of the time). Nine snaps can't tell you either; a few hundred can, and that is what the next chapter does with them.

💡 Watch for it on Sunday: the live run

For **one half** of one game, watching one offense:

- **Before every snap**, run the card in order: situation, personnel, formation, motion response, box, shell, leverage, pressure, call. Say the call and a confidence between 50% and 90% out loud *before* the snap. If the broadcast is still on a replay, call it from the situation and write “sit”.
- **On the snaps where you have time**, add one second-layer call with its own confidence: on likely passes, the coverage family or the rush (four, or five or more); on likely runs, the direction.
- **During the replay**, write the line: down, to-go, pers, align, call, conf, clue, and the second-layer call if you made one. Fill in the result.
- **At each stoppage**, update the tally of what this team does from each look today, and note anything it changed.
- **When the offense goes no-huddle**, switch to the short checklist (1, 3, 6, 9) and lower your second-layer confidences.
- **At halftime**, score it: hit rate, average confidence, and the hit rate of your 80%-plus calls. Keep the sheet. Film Study adds a second half of work to it: the All-22, the chart, and the score.

Predict the play

💡 Predict the play 1: first down, two backs

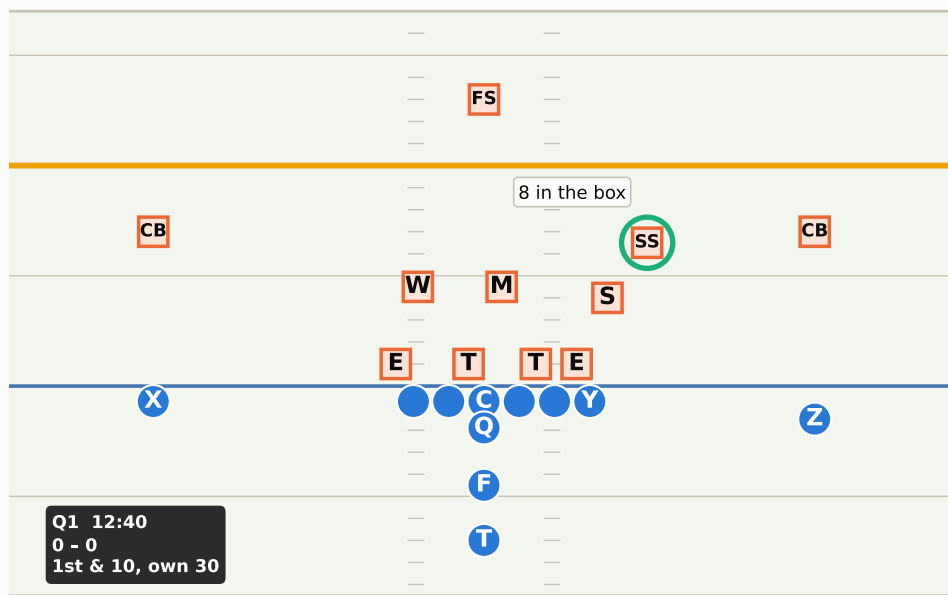


Figure 13: First quarter, 0–0. First-and-10 at the offense’s own 30. The offense has 21 personnel (two backs, one tight end) in an I-formation: quarterback under center, fullback (F) and tailback (T) stacked behind him, the tight end (Y) on the right. The defense is in base with the strong safety (ringed) walked down to the edge of the box: eight defenders in the box, one safety deep in the middle.

Run or pass, how sure are you, and if it’s a pass, what kind?

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: the mugs on third down

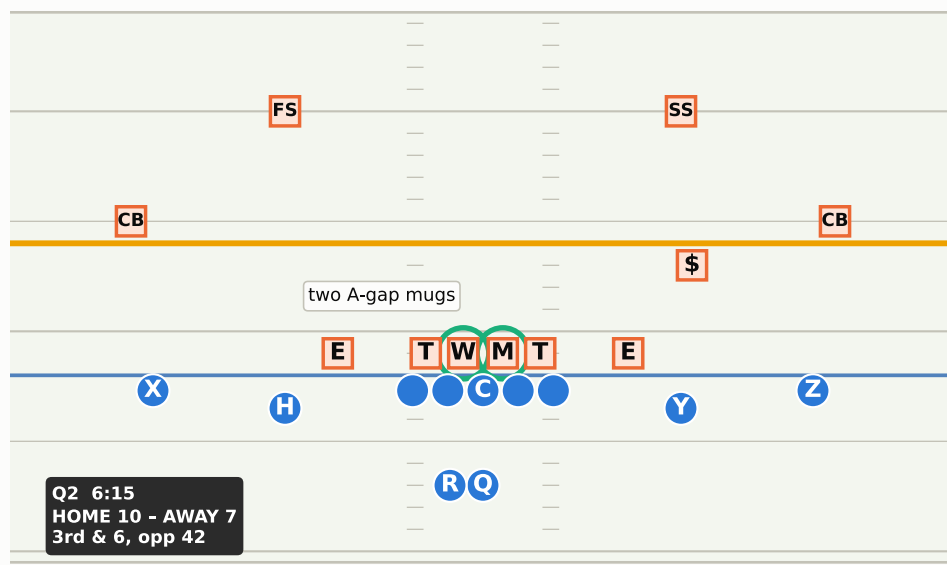


Figure 14: Third-and-6 at the opponent's 42, second quarter. Shotgun, 11 personnel, two receivers each side, the back to the quarterback's left. The defense shows a pressure look: both inside linebackers (ringed) have walked up into the A gaps, either side of the center, so six defenders are at the line. Behind them, two safeties stand 12 yards deep and both corners are seven yards off.

It's a pass; how many will rush, and where should the ball go?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: tempo after a first down

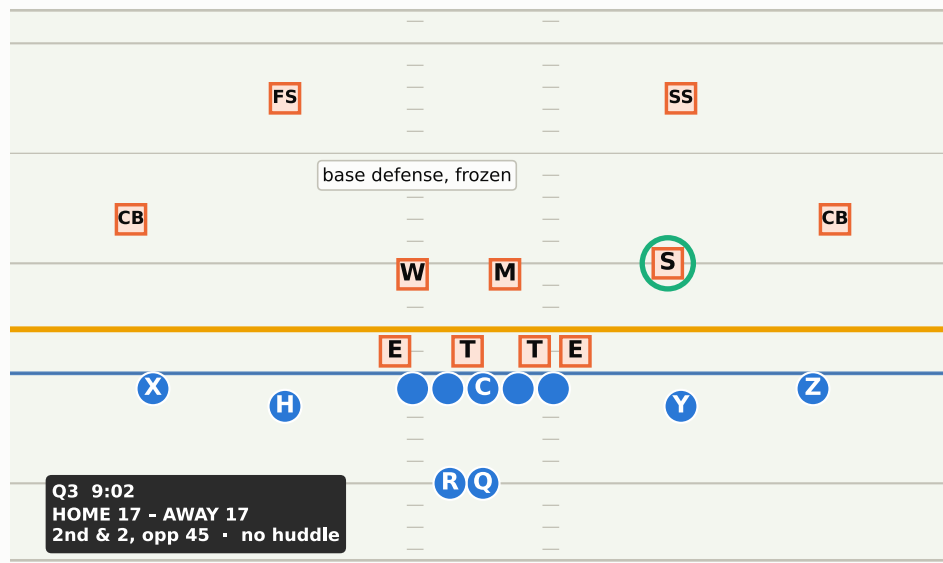


Figure 15: Second-and-2 at the opponent’s 45, third quarter. The offense just ran for 8 yards on first down out of 11 personnel and sprinted to the line without huddling or substituting. The defense, which had its base personnel on the field (three linebackers, two corners, two safeties), couldn’t substitute: the strongside linebacker (ringed) has had to walk out over the slot receiver on the right.

What is the offense hunting, and what’s your call?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **Order beats speed.** The master checklist asks nine questions in the order the game answers them (situation, personnel, formation, motion response, box, shell, leverage, pressure, call), and each answer changes what the next clue means.
- **Run or pass is mostly settled by step 3.** In 2024–2025 data the situation and the formation did almost all the work; motion, tempo and the box added under a point. Spend the later steps on the second layer: coverage, rush, where the ball goes.
- **Tells are leans with reasons.** Every tell in the catalogue leaks because of physics or assignment, and every one is broken on purpose by somebody. Trust them in proportion to how often they’re right, and expect the break where it’s cheapest for the offense.
- **Say a number.** A stated confidence between 50% and 90%, made before the snap, turns watching into something you can score. Start from the base rate, move with strong clues, and rarely go past 90.
- **Score the read, not the play.** A correct call on a failed play is a correct call, and a 65% call that misses on a tendency-breaker is honest reading.

Connections

Builds on: every “checklist so far” card (Motion and Shifts, Offensive Systems, Disguise and Rotation, Zone Blitzes); the first tells and the prediction log in How to Watch a Broadcast; tempo and in-game adjustment in In-Game Adjustments and Tempo; the clock in The Clock, the Two-Minute Drill, and Fourth Down; base rates and tendencies in Tendencies: PROE, Personnel, Motion, and Coverage.

Sets up: Film Study: All-22, Charting a Game, and Scoring Yourself, which takes the log you keep live, adds the All-22 film and a full chart of the game, and scores your season of stated confidences with a Brier score and a calibration plot.

Case studies: the film-room drive was run by an offense from the Shanahan family (Shanahan’s Wide Zone Tree) against a defense from the two-high era (Fangio’s Two-High Revolution and Macdonald’s Answer).

i Go deeper

- Atul Gawande, *The Checklist Manifesto* (2009): why short, ordered checklists help experts rather than replace them.
- Philip Tetlock and Dan Gardner, *Superforecasting* (2015): stating probabilities, keeping score and updating, the habits behind this chapter's confidence column.
- Chris B. Brown, *The Art of Smart Football* (2015), and his Smart Football site: the pre-snap game from the play-caller's side.
- Cody Alexander's MatchQuarters newsletter and Ted Nguyen's film breakdowns at *The Athletic*: coverage and pressure tells explained by people who watch the All-22 every week.
- The nflverse project (nflready) and FTN's charting: everything in this chapter's charts, free to download, and the raw material for your own base rates.

Answers to Predict the play

Answer 1.

Run, about 70%; and if it's a pass, expect play-action. *Situation:* first-and-10 is a coin flip (50% passes in 2025). *Personnel and formation:* 21 personnel with the quarterback under center moves it a long way: on first-and-10 from that look, offenses threw only 31% of the time in 2025. *Box:* eight in the box with one safety deep is the defense loading up against the run, and offenses facing it from this look threw a little less often (27%, though that's only 77 snaps).

So why not 85 or 90? Because this is the exact picture play-action is built for. One deep safety and eight defenders committed to the run leave the deep middle and the space behind the linebackers thin, and an offense that gets the linebackers to step up for the fake can throw a post or a deep crossing route behind them (Play-Action, Bootlegs and Screens). The I-formation also hides the run's direction: with the fullback straight behind the quarterback there's no offset to read. A good log line: *call R, conf 0.70, clue align, l2 "if pass, PA shot"*.

Answer 2.

Pass, near certain; four rushers is still the best single call (about 60%), and the ball should come out fast. Run or pass is the cheap part: on third-and-4 to 6 from the shotgun with 11 personnel, offenses threw 93% of the time in 2025. Spend the rest on step 8.

Count the deep players first. Two safeties at 12 yards and two corners seven yards off are four defenders who are, for now, playing coverage. That leaves seven for the rush and the underneath zones, and six of them are standing at the line. If all six rushed, there would be one defender underneath, which almost no defense risks with the ball at midfield. So the realistic menu is: both mugs drop and four rush (a simulated pressure, the creeper of Zone Blitzes); one mug comes and one drops; or a five-man pressure with a safety rotating down. On third-and-4 to 6 in 2025, defenses sent five or more on only 27% of dropbacks. A pressure look moves that number up, but probably not past half, so "four rush, 60%" is an honest call, and "five or more, 40%" is the alternative you keep in mind.

Where the ball goes: the offense has to protect against six, so the quarterback will likely set the protection to the mugs and have a quick answer built in. Expect a throw in under 2.5 seconds, near the sticks, to the side a dropping linebacker has the farthest to run to. And if a mug drops into the throwing lane, that is the zone blitz doing its job.

Answer 3.

A lean to pass, about 60%, aimed at the linebacker on the slot. *Situation:* second-and-2 is a run down overall (only 31% passes in 2025), but from the shotgun with 11 personnel it was 57%, and with no huddle 64% (a small sample, 131 snaps, so don't lean on the decimal). Second-and-short is also the down where offenses take shots, because a failure still leaves third-and-short.

Personnel and tempo: this is the substitution freeze at work. Because the offense didn't substitute, the defense had no guaranteed chance to swap a linebacker for a nickel back, so its strongside linebacker is now covering a slot receiver in space. That is the matchup the offense sprinted to the line to get: a quick throw or a crossing route at the linebacker, or a deeper shot if the safeties cheat up for the run.

Why only 60%: the same frozen base defense has only six in the box, and with the tight end split out as a slot the offense has five linemen to block them; a shotgun run in which the quarterback reads one defender instead of blocking him evens that count. A team that just gained eight on the ground may simply run it again. Tempo tells you the offense likes something about this picture; it doesn't tell you which of the two things it likes. A good log line: *call P, conf 0.60, clue tempo, l2 "target the slot", 0.55*.